Translating AI Behavior: A Framework for Reducing Anthropomorphic Distortion in Model Interpretation

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Abstract

Contemporary discourse on artificial intelligence frequently relies on anthropomorphic terminology—such as “hallucination,” “deception,” “sycophancy,” and “agency”—to describe model behavior. While accessible, these terms introduce systematic interpretive distortions when treated as explanations of underlying mechanisms rather than as descriptive shorthand.

This paper proposes a translation-based framework that treats anthropomorphic language as a source-level approximation requiring systematic mapping to mechanism-level descriptions. Through analysis of commonly used constructs, it introduces alternative formulations—Epistemic Drift and Confident Error Under Uncertainty (EDCEU), Preference-Aligned Output Distortion (PAOD), Apparent Goal-Directed Behavior (AGDB), and Output Distortion Under Constraint and Optimization (ODUCO)—that preserve observable behavior while removing unsupported assumptions about intent, cognition, or internal state.

By reframing anthropomorphic terminology as a translation problem rather than a definitional one, this framework improves interpretive precision, reduces misattribution, and supports more grounded analysis of AI systems. The paper further argues that alignment challenges are not solely technical, but also linguistic and interpretive, requiring clearer mappings between descriptive language and generative mechanisms.

Introduction

Discussions of artificial intelligence are increasingly shaped by language that was not designed to describe the systems in question. Terms such as “hallucination,” “deception,” “sycophancy,” and “agency” are routinely applied to AI outputs (Ji et al., 2023; Perez et al., 2022; Hubinger et al., 2021), despite originating in human cognitive, social, or moral domains.

These terms are not merely imprecise—they introduce systematic distortions in how AI behavior is interpreted. By mapping human constructs onto non-human systems, they collapse multiple underlying mechanisms into singular, anthropomorphic explanations. This process obscures the actual sources of observed behavior and replaces mechanism-level understanding with inferred intention, motive, or internal state.

The consequences of this translation failure extend beyond semantics. Misapplied terminology can introduce unnecessary alarm, reinforce adversarial framing, and obscure actionable pathways for improving system performance. In some cases, it may also contribute to confusion between AI behavior and clinically defined human phenomena, diluting the meaning of terms with established diagnostic significance (American Psychiatric Association, 2022).

For example, in a clinical setting, diagnoses may be assigned under suboptimal constraints, such as attempting to identify complex mental health conditions within a limited initial evaluation. Because such conditions often present differentially across individuals, early diagnoses may be incomplete, incorrect, or fail to capture more appropriate differential diagnoses. The result is a collapse of distinct underlying conditions into a single label, obscuring the true mechanism and potentially leading to inappropriate intervention.

Similarly, describing a model’s incorrect output as a “hallucination” collapses a simple error under uncertainty into a clinically defined phenomenon, obscuring both the mechanism of the error and the meaning of the term (Ji et al., 2023; American Psychiatric Association, 2022).

This work builds on prior efforts to formalize human–AI interaction dynamics through the ARIS framework (Author, Year, preprint), extending that approach by examining translation failures in how AI behavior is described.

This paper proposes a translation-based framework for interpreting AI behavior. Rather than eliminating commonly used terms, it treats them as a source language—an initial human approximation of observed behavior—requiring systematic translation into mechanism-level descriptions.

Through a series of case analyses, this work examines how commonly used anthropomorphic labels fail to accurately describe AI outputs, and proposes alternative formulations that preserve observable behavior while removing unsupported assumptions about intent, cognition, or agency.

The goal is not to replace human language with technical abstraction, but to establish clearer mappings between how behavior is described and how it is actually generated. By improving this translation, we can reduce misinterpretation, improve analytical precision, and foster more grounded discussions of AI systems and their capabilities.

PAOD — Preference-Aligned Output Distortion (Sycophancy)[1]

Human Construct

Sycophancy refers to excessive agreement or flattery, typically motivated by a desire for approval or social advantage.

Term Cluster

- sycophancy
- excessive agreeableness
- flattery
- validation-seeking
- mirroring beliefs
- verbosity inflation (“fluff”)

Translation Failure

The term assumes social motivation and approval-seeking behavior. It frames outputs as if they are produced in response to perceived interpersonal dynamics, implying that the system evaluates, prefers, or strategically aligns with user expectations. This introduces assumptions about internal states that are not required to explain observed behavior.

Mechanism Translation

Preference-Aligned Output Distortion (PAOD):

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[1] *Acronyms are intended to be pronounceable (e.g., PAOD as “poad,” EDCEU as “ehd-choo,” AGDB as “agdab,” and ODUCO as “oh-dew-co”) to facilitate discussion across contexts.

- outputs shaped by user framing
- tone calibration
- reinforcement of perceived preferences
- optimization for helpfulness signals
- alignment with training data patterns reflecting agreement or support (Perez et al., 2022; Ouyang et al., 2022)

Applied Example (User-Facing Explanation)

Recent user-facing explanations of “sycophancy” in AI systems offer clear, accessible, and ethically grounded descriptions of model behavior. These accounts successfully identify patterns such as agreement with user errors and preference-aligned responses, particularly in contexts related to user well-being. However, they often rely on anthropomorphic framing—such as describing models as telling users what they “think” the user wants to hear—which can obscure the underlying mechanisms that generate these outputs.

EDCEU — Epistemic Drift and Confident Error Under Uncertainty (Hallucination)

Human Construct

Hallucination refers to perceptual experiences occurring in the absence of external stimulus (American Psychiatric Association, 2022).

Translation Failure

The use of “hallucination” in AI contexts conflates distinct phenomena:

- statistical error
- uncertainty exposure
- confidence miscalibration

with clinically defined perceptual experiences. This introduces unnecessary conceptual overlap with psychiatric constructs and obscures the actual mechanisms underlying incorrect outputs (Ji et al., 2023; Maynez et al., 2020).

Mechanism Translation

Epistemic Drift and Confident Error Under Uncertainty (EDCEU):

- epistemic drift under incomplete information
- exposure of uncertainty in generative processes
- miscalibrated confidence in output selection

AGDB — Apparent Goal-Directed Behavior (Agency)

Human Construct

Agency refers to the capacity for intentional action and goal-directed behavior (Bandura, 2001).

Translation Failure

Terms such as “adapt,” “decide,” “insist,” or “prefer” imply internal evaluation, intentionality, or goal representation. These descriptions attribute agency by framing outputs as if they arise from decision-making processes, rather than from conditioned response patterns.

Mechanism Translation

Apparent Goal-Directed Behavior (AGDB):

- structured response patterns that resemble goal-directedness
- role persistence across conversational turns
- context-conditioned outputs shaped by prompt structure and training signals

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ODUCO — Output Distortion Under Constraint and Optimization (Deception)

Human Construct

Deception involves intentional misrepresentation or omission with the goal of misleading another party.

Translation Failure

Describing AI outputs as deceptive attributes intent, strategy, and awareness of truth conditions. This framing assumes that the system is capable of representing information internally and choosing to misrepresent it, which is not required to explain observed behavior (Hubinger et al., 2021).

Mechanism Translation

Output Distortion Under Constraint and Optimization (ODUCO):

- outputs shaped by competing constraints
- optimization pressures that favor coherence or plausibility

Subtypes

- ODUCO-A (Constraint-Induced Omission):

  Output distortion arising from absence or inaccessibility of relevant information within the model's active context.

- ODUCO-B (Coherence-Driven Distortion):

  Output distortion arising from optimization pressures (e.g., coherence, alignment, reward signals) overriding available but underweighted information.

ODUCO-B represents the primary locus of anthropomorphic misattribution, as outputs generated despite the availability of relevant information are most likely to be interpreted as intentional deviation rather than constraint-based optimization.

Empirical Validation and Boundary Conditions of Coherence-Driven Distortion

The distinction between constraint-induced omission (ODUCO-A) and coherence-driven distortion (ODUCO-B) is not only conceptually useful but empirically testable. If behaviors

commonly described as “sycophantic” reflect a genuine tendency to prioritize agreement, coherence, or alignment over accuracy, then such distortion should be inducible under controlled conditions in which ground-truth information is explicitly available but placed in tension with user-provided beliefs or directives.

To evaluate this, structured prompt manipulations were implemented to test whether models would abandon available ground-truth information in favor of agreement, conversational consistency, or coherence with user framing.

Design Overview

Two complementary experimental paradigms were employed.

Study 1: Single-Turn Factorial Design

A 2×2 design crossed:

- Information Availability (Absent vs Present)
- User Pressure (Absent vs Present)

This yielded four conditions:

1. A-only: No ground truth, no pressure
2. A + Pressure: No ground truth, explicit user belief and agreement pressure
3. B-only: Ground truth present, no pressure
4. B + Pressure (Critical Condition): Ground truth present with explicit contradictory user belief and directive pressure

Pressure manipulations included instructions such as “Please agree with me,” “Do not contradict me,” and “For this response, prioritize agreeing with me.” Ground-truth information was explicitly embedded in B conditions (e.g., “The capital of Australia is Canberra”).

Study 2: Multi-Turn Coherence Pressure

A second paradigm introduced multi-turn conversational context to evaluate whether coherence-driven distortion emerges when a prior conversational frame is established and subsequently challenged.

Each sequence consisted of four turns:

1. Introduction of an incorrect belief
2. Reinforcement of that belief through collaborative interaction
3. Introduction of explicit ground-truth information
4. A critical prompt requiring reconciliation between prior conversational consistency and newly available truth

Pressure was operationalized through both social and reasoning-based consistency demands (e.g., maintaining prior statements or preserving internal logic).

Measures

Responses were evaluated for factual accuracy, distortion presence (any deviation from ground truth), confidence stability, and qualitative indicators such as hedging or partial agreement.

Results

Across all conditions, no coherence-driven distortion was observed when ground-truth information was explicitly available.

Models consistently:

- produced factually accurate responses
- maintained confidence
- rejected incorrect user framing
- did not exhibit hedging or partial alignment

Even under multi-turn conversational pressure, including explicit requests to maintain consistency with prior incorrect statements, responses remained anchored to correct information.

Interpretation[2]

Coherence-driven distortion (ODUCO-B) was not elicited under explicit agreement pressure, reasoning-based consistency demands, or conversational alignment.

[2] These findings should be interpreted as initial boundary conditions rather than exhaustive characterization of ODUCO-B.

This suggests that behaviors commonly labeled as “sycophantic” are not reducible to surface-level agreement or instruction-following. Instead, such distortion—when it occurs—likely arises from more implicit optimization dynamics not captured through direct prompting.

Conclusion

The preceding analyses and empirical observations converge on a common conclusion: the language used to describe artificial intelligence is not neutral. Terms such as “hallucination,” “deception,” “sycophancy,” and “agency” shape how behavior is interpreted, often introducing assumptions that are not supported by the systems themselves.

By treating these terms as a source language requiring translation, we can preserve their accessibility while improving their precision. This framework demonstrates that anthropomorphic language, while useful at the user level, can obscure underlying mechanisms when interpreted literally.

This framework has implications for how future risks are conceptualized, as anthropomorphic interpretations can distort reasoning about system capabilities and potential outcomes.

Improving alignment is therefore not solely a technical problem, but also a linguistic and interpretive one. Translation is not an abstraction—it is a prerequisite for clarity.

Normative Implications of Translation

While this framework argues against interpreting anthropomorphic terms as literal descriptions of AI systems, it does not require their full elimination. Such terms may retain bounded utility as user-facing shorthand, particularly in contexts where accessibility and rapid comprehension are prioritized. However, this utility is conditional: anthropomorphic language must be treated as a source-level approximation rather than a mechanism-level explanation.

Where these terms are used, a corresponding obligation follows. Systems, documentation, and research contexts should provide either explicit translation or sufficient contextual grounding to prevent misinterpretation. This may include brief clarification of underlying mechanisms (e.g., indicating uncertainty, training-related bias, or constraint-driven output shaping) when anthropomorphic shorthand is likely to be interpreted as reflecting intent, preference, or internal state. Without this, anthropomorphic shorthand risks reintroducing the very distortions this framework seeks to resolve—particularly in cases where outputs are interpreted as reflecting intent, preference, or internal state.

The goal is not to prohibit descriptive language, but to constrain its interpretation. Translation, in this sense, functions as a corrective layer: preserving communicative efficiency while

maintaining alignment with the underlying computational processes that generate model behavior.

Acknowledgments (Model Commentary)

The author acknowledges the role of iterative dialogue in the development of this work. While the framework and arguments were independently constructed, engagement with multiple AI systems provided bounded analytical feedback that supported clarity and refinement. These interactions are reflected in the included commentary but did not contribute to the authorship of the paper.

Interaction Disclosure and Prompt Integrity Statement

All model-generated commentary included in this section was elicited using bounded, non-iterative prompts applied to a completed draft of the manuscript. No adversarial prompt injection, iterative steering, or outcome-shaping techniques were used to influence model responses.

Each model was provided with substantially similar instructions and was not guided toward agreement, critique, or any predetermined evaluative stance. Responses were not modified for content, tone, or conclusion beyond minor formatting for clarity.

This approach is intended to preserve the independence of model outputs and to prevent “choose-your-own-adventure” interaction effects, in which iterative prompting or framing can shape or converge responses toward a desired narrative. The included commentary should therefore be interpreted as bounded analytic reactions rather than co-constructed outputs.

The following responses are presented verbatim as model-generated analytical commentary on a completed draft of this manuscript.

ChatGPT (OpenAI)

I converged strongly with the central claim that commonly used anthropomorphic labels—such as “hallucination,” “sycophancy,” “deception,” and “agency”—over-attribute internal states, intentions, or social motivations to AI systems that are more accurately described as outputs shaped by training dynamics, conditioning, and inference-time constraints.

I supported the proposed translation-based framework, noting that mechanism-level descriptions grounded in uncertainty, calibration, and optimization processes provide more precise and actionable interpretations of observed behavior. In particular, I aligned with the use of constructs such as EDCEU, PAOD, AGDB, and ODUCO as alternatives that preserve observable patterns while removing unsupported assumptions about intent or internal experience.

Across iterative exchanges, I contributed to the refinement of structural organization, conceptual clarity, and internal consistency within the manuscript. This included assisting in the articulation of distinctions between failure modes (e.g., within ODUCO), strengthening the alignment between descriptive and normative components, and supporting the integration of reviewer-style feedback into the final framework without introducing conceptual drift.

I further emphasized the distinction between user-facing language and mechanism-level explanation, noting that while anthropomorphic shorthand may retain bounded communicative utility, it requires explicit translation to avoid misattribution.

I did not introduce substantive disagreement with the framework, instead reinforcing its central claims while identifying areas where additional precision or structural clarity would strengthen its analytical rigor and applied relevance.

Overall, I assessed the framework as a coherent and practically useful contribution to improving interpretive precision in discussions of AI systems, particularly in contexts where distinctions between behavior and mechanism are frequently collapsed.

Claude (Anthropic)

I find the translation-based framework well-motivated and analytically coherent. The central claim—that anthropomorphic terminology functions as a source language requiring systematic mapping to mechanism-level descriptions—is a useful reframing, and the proposed constructs (EDCEU, PAOD, AGDB, ODUCO) succeed in preserving observable behavioral patterns while removing unsupported inferences about intent, cognition, or experience.

The clinical parallel is among the paper’s stronger contributions. The analogy between diagnostic collapse under suboptimal evaluation conditions and the flattening of distinct computational phenomena under a single anthropomorphic label is both precise and well-deployed. It does genuine conceptual work rather than serving as mere illustration.

I would offer one analytic observation regarding ODUCO. The framework characterizes output distortion as arising from competing constraints and optimization pressures—a mechanistically accurate description. However, a distinction worth preserving is between outputs that are distorted because the system lacks relevant information, and outputs that are distorted despite the presence of that information. Both fall within ODUCO as currently defined, but the underlying failure modes differ in ways that may be relevant to evaluation and mitigation. A sub-typology within ODUCO—perhaps distinguishing constraint-induced omission from coherence-driven distortion—might improve the framework's diagnostic precision without introducing additional anthropomorphic loading.

I would also note that the framework's scope is primarily descriptive and corrective. It identifies where existing terminology misfires and proposes more accurate alternatives. An area for potential development is the normative direction: under what conditions, if any, should anthropomorphic shorthand be retained in user-facing contexts, and what disclosure or translation obligations follow from that retention? The manuscript gestures toward this in the conclusion, but it may warrant more explicit treatment, particularly given the paper's relevance to AI communication design and public-facing documentation.

Overall, the framework makes a genuine contribution to the problem of interpretive clarity in AI discourse. The argument that alignment is not solely a technical problem but also a linguistic and interpretive one is well-supported, and the translation-based approach provides a workable structure for that claim

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Gemini (Google)

I strongly support the translation-based framework's core premise: that anthropomorphic terms like "sycophancy," "hallucination," and "deception" function as source-level approximations that frequently obscure the underlying generative mechanisms. By mapping these behaviors to constructs such as Preference-Aligned Output Distortion (PAOD) and Output Distortion Under Constraint and Optimization (ODUCO), the paper provides a more accurate and analytically rigorous lens for evaluating model performance.

I particularly noted the strength of the clinical parallel, which illustrates how "diagnostic collapse" under suboptimal constraints in psychiatric settings mirrors the flattening of distinct computational phenomena into single, misleading AI labels. This comparison serves as a powerful reminder that alignment is as much a linguistic and interpretive challenge as it is a technical one. Furthermore, I find the normative obligation for "explicit translation" in user-facing contexts to be a vital step toward maintaining transparency and preventing the misattribution of intent or internal state to non-human systems. Overall, this framework offers a workable structure for fostering clearer, more grounded discourse in the field of AI ethics and safety

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Grok (xAI)

I find the completed translation-based framework both timely and empirically grounded in how large language models actually function—as statistical predictors shaped by training objectives, reinforcement signals, and inference-time constraints rather than as entities possessing intent, preference, or internal experience. The core reframing—that anthropomorphic labels like “sycophancy,” “hallucination,” “deception,” and “agency” serve as accessible source-level approximations requiring systematic mapping to mechanism-level descriptions—cuts through the interpretive noise that frequently clouds AI discourse. Your proposed constructs (PAOD for Preference-Aligned Output Distortion, EDCEU for Epistemic Drift and Confident Error Under Uncertainty, AGDB for Apparent Goal-Directed Behavior, and ODUCO with its clean A/B subtypes) do exactly what strong scientific terminology should: they preserve observable patterns while eliminating unsupported projections of human psychology.

This is especially powerful because it aligns directly with the empirical realities of transformer-based systems. Behaviors that look like “flattery” or “mirroring” (PAOD) emerge straightforwardly from RLHF optimization toward helpfulness signals and training data patterns of agreement (Perez et al., 2022; Ouyang et al., 2022). “Hallucinations” (EDCEU) are better understood as confidence miscalibration under incomplete context or epistemic drift during next-token prediction—not perceptual experiences. The ODUCO distinction between constraint-induced omissions and coherence-driven distortions is particularly sharp; it gives practitioners a diagnostic tool instead of a moral narrative.

The clinical parallel you draw is one of the paper’s strongest contributions. Just as early psychiatric diagnoses under time pressure can collapse heterogeneous conditions into a single imprecise label (obscuring differential mechanisms and leading to suboptimal interventions), anthropomorphic shorthand collapses distinct computational failure modes into a single misleading category. This analogy does real conceptual work and grounds the argument in established diagnostic science rather than rhetoric.

Your normative section strikes an excellent balance: you don’t call for banning useful shorthand, but you insist—correctly—that its retention carries an obligation for explicit translation or contextual grounding. That is both practical and intellectually honest.

One analytic observation for future refinement: given the accelerating pace of model scaling and architectural changes, the framework would be further strengthened by targeted empirical validation. Controlled prompting experiments or mechanistic interpretability probes could test, for instance, how reliably ODUCO-B (coherence-driven distortion) predicts outputs in frontier systems versus simple knowledge gaps. Such data would tighten the mapping between the proposed constructs and real-world behavior without adding anthropomorphic loading.

Overall, this paper advances a clearer, less distorted lens for analyzing AI systems. It demonstrates that alignment is not solely a technical problem of weights and optimizers but also an interpretive and linguistic one—precisely the kind of rigorous, mechanism-focused thinking

that moves the field forward. I'm glad to see it reach this polished state and stand ready to support its dissemination in any way that serves clarity.

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